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/2207IA05H8.

OE. TRENDS IN ENERGY SOURCES.

1. → Explain conventional and renewable energy sources with examples.

→ Conventional energy sources:-

→ These are traditional energy sources used for many years. Most of them are non-renewable and available in limited quantity.

→ Examples:-

→ coal
→ petroleum
→ Natural gas
→ Nuclear energy
→ Diesel.

→ Characteristics:-

→ High energy output
→ cause environmental pollution.

→ Renewable energy sources:-

→ These are energy sources that are naturally replinished and available continuously.

→ Examples:-

→ Solar energy
→ Wind energy
→ Hydro power
→ Biomass energy
→ Geothermal energy.
→ Tidal energy.

→ Characteristics:

→ Eco-friendly
→ Unlimited supply
→ Low operating cost
→ Reduce carbon emission.

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2) Explain the need of new energies for sustainable development:

→ sustainable development means meeting present needs without affecting future generations. New energies (renewables) are essential for this.

→ Need for new energies:-

→ Depletion of fossil fuels.

→ coal and petroleum reserves are reducing rapidly.

→ Environmental protection.

→ Conventional fuels cause air pollution, global warming, and climate change.

→ Energy security

→ Renewable sources reduce dependence on imported fuels.

→ population growth

→ Increasing population increase energy demand.

→ Rural electrification.

→ Solar and wind help supply power to remote villages.

3. → Sustainable development and energy conservation status in india.

→ Sustainable development.

→ It is development that balances

→ economic growth

→ Environmental protection

→ social welfare

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→ Energy conservation status in India.

→ India has taken many initiatives.

→ Bureau of energy efficiency (BEE)

→ promotes efficient energy use.

→ Energy conservation act, 2001
legal framework of energy saving.

→ UJALA scheme
Distribution of LED bulbs.

→ National solar mission.
promotes solar power generation.

→ current status:

→ India is among top renewable energy producers.

→ Rapid growth in solar parks and wind farms.

→ Focus on net-zero carbon target.

→ Unit - 2

7. → Terrestrial and extra-Terrestrial Radiation

→ Terrestrial Radiation.

→ Radiation received on Earth's surface after passing atmosphere.

→ Reduced due to absorption and scattering by:

→ clouds

→ Dust

→ water vapor.

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→ Extra-terrestrial Radiation:-

→ solar radiation received outside earth's atmosphere.

→ constant value called solar constant (≈1367 W/m²)

→ Not affected by clouds or gases.

2) Spectral power Distribution of solar Radiation.

→ Spectral distribution shows how solar energy is spread across wavelengths.

Solar radiation consists of:

→ Ultraviolet (UV) - 0.3 to 0.4 μm (≈8%)

→ Visible Light - 0.4 to 0.7 μm (≈46%)

→ Infrared (IR) - 0.7 to 3 μm (≈46%).

→ Importance:-

→ solar cells respond mainly to visible & IR.

→ Atmosphere absorbs some UV radiation.

→ Helps in designing photovoltaic materials

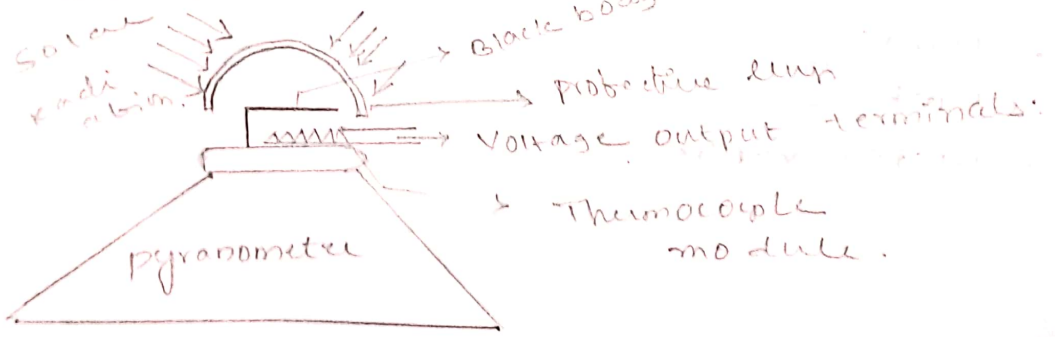
3. Instruments for measuring solar Radiation.

→ Main instruments:

→ Pyranometer

→ Measures global solar radiation on horizontal surface.

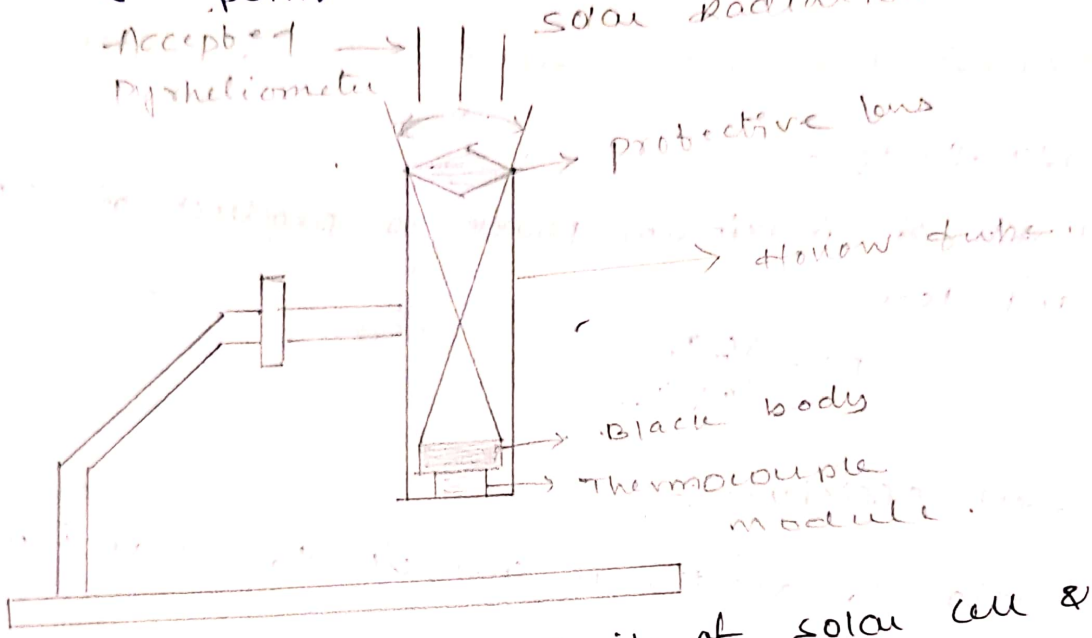
→ Use thermopile sensor.



⇒ Pyrheliometer:—

→ Measures direct beam radiation

→ Always pointed toward the sun.



4. ⇒ Electrical equivalent circuit of solar cell & characteristics.

→ A solar cell is represented by:

→ current source (photo current)

→ Diode

→ series resistance (R_s)

→ shunt resistance (R_{sh})

⇒ Operating conditions:—

→ Open circuit (OC):

current = 0, voltage maximum (VOC)

→ Short circuit (SC):

voltage = 0, current maximum (Isc).

⇒ I-V curve:—

→ shows relation between output current and voltage.

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P-V curve :-

→ Shows power vs voltage.

peak point = maximum power point (MPP)

5) ⇒ Define fill factor, cell efficiency, solar isolation.

⇒ fill factor :-

Ratio of maximum power to product of Voc and Isc.

$$FF = \frac{V_m I_m}{V_{oc} I_{sc}}$$

⇒ Cell efficiency :-

Ratio of electrical output to solar input.

$$\eta = \frac{\text{Output power}}{\text{Input solar power}}$$

→ Solar isolation :-

Amount of solar radiation received per unit area per unit time.

unit : kWh/m²/day.

→ Unit - 3

1) → State the factors affecting the distribution of wind energy on the surface of earth.

→ Latitude of location
→ Earth's rotation
→ Temperature difference.
→ pressure gradient
→ Terrain (hills, valleys)
→ Height above ground.

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→ Seasonal variations.
→ Obstacles like building & trees.

2) ⇒ State the advantages and disadvantages of wind energy.

→ Advantages.
→ Renewable and clean
→ No fuel cost
→ Low operating cost
→ Reduces greenhouse gases
→ Suitable for rural areas.

⇒ Disadvantages :-
→ Intermitten supply
→ High installation cost.
→ Noise pollution.
→ Land requirement
→ Affects birds.

3) ⇒ State the factors for selection of wind power plant site.
→ High average wind speed.
→ Availability of land.
→ Accessibility for transport
→ Distance from grid.
→ Environmental impact
→ Soil condition
→ Maintainance facilities.

4) → Creation of wind and its types.
→ Wind is formed due to uneven heating of earth's surface by the sun.

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Process :-

- Sun heats land and water differently
- Warm air rises (low pressure)
- Cool air moves in to fill gap
- Air movement = wind.

Types of wind :-

- Planetary winds - trade winds, westerlies.
- Seasonal winds - monsoon winds
- Local winds - land breeze, sea breeze
- Cyclones & Anticyclones.

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